

	A→C	A→D	A→W	C→A	C→D	C→W	D→A	D→C	D→W	W→A	W→C	W→D	AVG
NA	41.7	44.6	31.9	53.1	47.8	41.7	26.2	26.4	52.5	27.6	21.2	78.3	41.1
SA	37.4	36.3	39.0	44.9	39.5	41.0	32.9	34.3	65.1	34.4	31.0	62.4	41.5
GFK	41.9	41.4	41.4	56.0	42.7	45.1	38.7	36.5	74.6	31.9	27.5	79.6	46.4
TCA	35.2	39.5	29.5	46.8	52.2	38.6	36.2	30.1	71.2	32.2	27.9	74.5	42.8
CORAL	45.1	39.5	44.4	52.1	45.9	46.4	37.7	33.8	84.7	36.0	33.7	86.6	48.8

Table 3: Object recognition accuracies of all 12 domain shifts on the Office-Caltech10 dataset (?) with SURF features, using the “full training” protocol.

	C→I	C→S	I→C	I→S	S→C	S→I	AVG
NA	66.1	21.9	73.8	22.4	24.6	22.4	38.5
SA	43.7	13.9	52.0	15.1	15.8	14.3	25.8
GFK	52	18.6	58.5	20.1	21.1	17.4	31.3
TCA	48.6	15.6	54.0	14.8	14.6	12.0	26.6
CORAL	66.2	22.9	74.7	25.4	26.9	25.2	40.2

Table 4: Object recognition accuracies of all 6 domain shifts on the Testbed Cross-Dataset (?) dataset with DECAF-fc7 features, using the “full training” protocol.

	K→D	D→B	B→E	E→K	AVG
NA	72.2	76.9	74.7	82.8	76.7
TCA	60.4	61.4	61.3	68.7	63.0
SA	78.4	74.7	75.6	79.3	77.0
GFS	67.9	68.6	66.9	75.1	69.6
GFK	69.0	71.3	68.4	78.2	71.7
SCL	72.8	76.2	75.0	82.9	76.7
KMM	72.2	78.6	76.9	83.5	77.8
CORAL	73.9	78.3	76.3	83.6	78.0

Table 5: Review classification accuracies of the 4 standard domain shifts (?) on the Amazon dataset (?) with bag-of-words features.

Amazon-SURF). Similarly, the improvement on images (Tables 1-4) is much larger than text (Table 5), possibly because bag-of-words text features are extremely sparse and less correlated than image features. As demonstrated in (?), high level deep features are more “parts” or “objects”. Intuitively, “parts” or “objects” should be more strongly correlated than “edges” (e.g., arm and head of a person are more likely to appear jointly).

These findings suggest that CORAL is extremely valuable in the era of deep learning. Applying CORAL to deep text features is part of future work.

6 Conclusion

In this article, we proposed a simple, efficient and effective method for domain adaptation. The method is “frustratingly easy” to implement: the only computation involved is re-coloring the whitened source features with the covariance of the target domain.

Extensive experiments on standard benchmarks demonstrate the superiority of our method over many existing state-of-the-art methods. These results confirm that CORAL is applicable to multiple features types, including highly-performing deep features, and to different tasks, including computer vision and natural language processing.

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